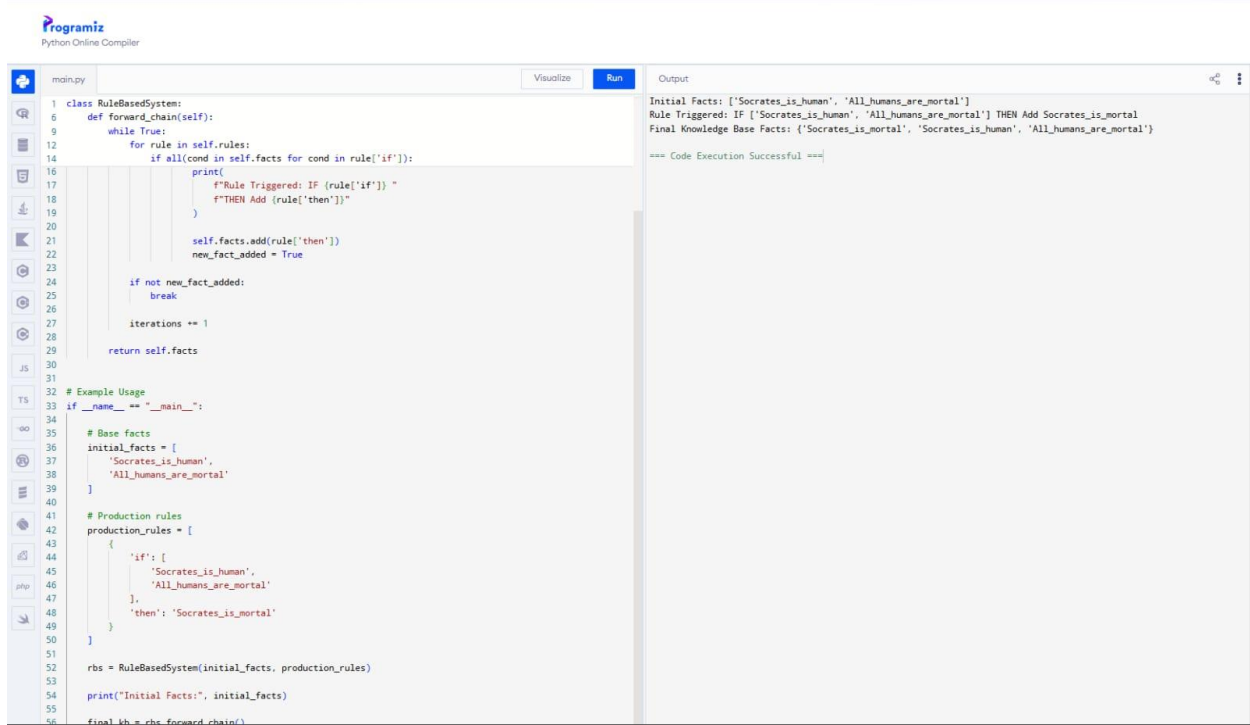


AKSHAY V

24BECS174

AIML PROGRAM 5



The screenshot displays a web-based Python IDE. The top bar includes the 'Programiz' logo and the text 'Python Online Compiler'. Below this, there are tabs for 'main.py', 'Visualize', and 'Run'. The 'main.py' tab is active, showing a Python script. The script defines a `RuleBasedSystem` class with a `forward_chain` method. This method iterates through a list of rules, checking if the conditions are met. If a rule's conditions are satisfied, it prints a message and adds the rule's conclusion to the facts. The script also includes an example usage section that initializes facts, defines production rules, and runs the forward chain.

```
1 class RuleBasedSystem:
2     def forward_chain(self):
3         while True:
4             for rule in self.rules:
5                 if all(cond in self.facts for cond in rule['if']):
6                     print(
7                         f"Rule Triggered: IF {rule['if']} "
8                         f"THEN Add {rule['then']}"
9                     )
10                    self.facts.add(rule['then'])
11                    new_fact_added = True
12
13                    if not new_fact_added:
14                        break
15
16                    iterations += 1
17
18            return self.facts
19
20 # Example Usage
21 if __name__ == "__main__":
22
23     # Base facts
24     initial_facts = [
25         'Socrates_is_human',
26         'All_humans_are_mortal'
27     ]
28
29     # Production rules
30     production_rules = [
31         {
32             'if': [
33                 'Socrates_is_human',
34                 'All_humans_are_mortal'
35             ],
36             'then': 'Socrates_is_mortal'
37         }
38     ]
39
40     rbs = RuleBasedSystem(initial_facts, production_rules)
41
42     print("Initial Facts:", initial_facts)
43
44     final_kb = rbs.forward_chain()
```

The 'Output' tab on the right shows the execution results:

```
Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal'] THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_mortal', 'Socrates_is_human', 'All_humans_are_mortal'}
=== Code Execution Successful ===
```